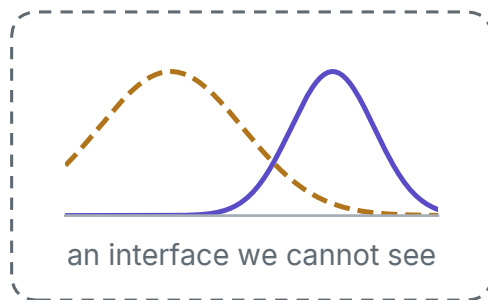
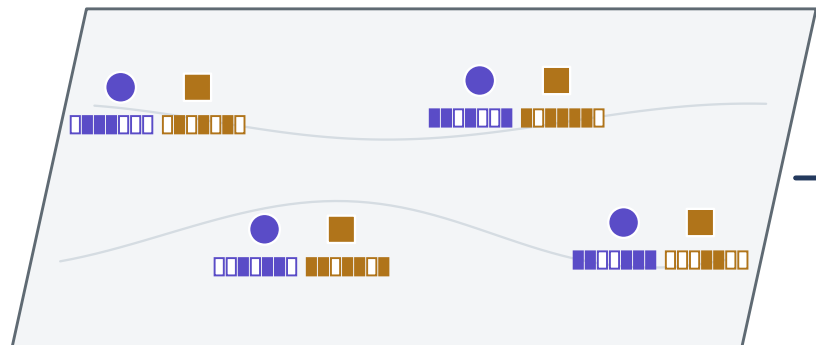


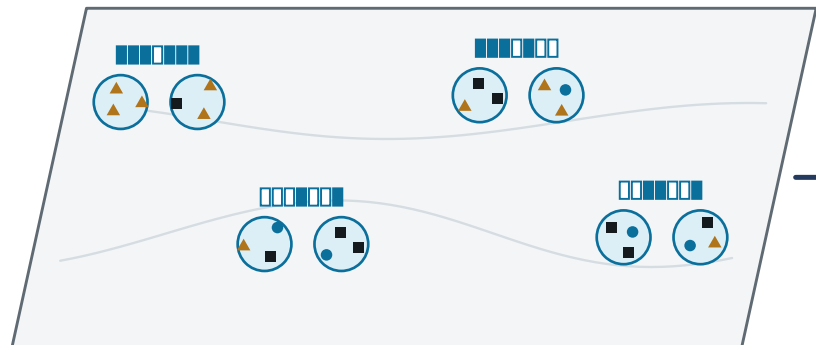
what we can sample, across places



**Are two species shaping
one another?
Where in their genomes?**

or is it shared environment,
dispersal, history?

what we can sample, across places



$$\boxed{g} + \bullet \blacksquare \blacktriangle + \boxed{g} \times \blacktriangle + \dots$$

genes, microbes and their
interactions shape host fitness



**Is the microbiome changing
host evolution?
Through what?**

or is it microbial ecology,
geography, demography?

What models, data and spatial replication make these mechanisms distinguishable?